

Ghassen Bargougui

Application Engineer · Fukuoka, Japan

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Profile

I am an Application Engineer with seven years of backend and full-stack experience across Tunisia, Luxembourg, and Japan. My work spans banking, insurance, and card payments, with a focus on Java, API design, and reliable delivery. Outside work, I build products from architecture to delivery: Slide Agent for editable AI-generated presentations, Pockito for personal and shared money management, and SubMate for subtitle translation. I speak Arabic natively, English and French fluently, and Japanese at a basic level.

Professional experience

Application Engineer

Dec 2025 - Present

Rakuten Card · Fukuoka, Japan

I work in Rakuten Card's Technology Incubation Team, helping lead hybrid-cloud migration with my manager and project manager across multiple teams while modernizing APIs.

- I upgraded the e-Navi API to JDK 25 and Helidon 4.3, resolving dependency and CDI issues, updating tests, and documenting the handover.
- I built a GitHub Actions pipeline for builds, Docker image publishing to Artifact Registry, and GKE deployments.
- I designed and validated Kubernetes deployments for JSF, API, and Engine modules, including services, environment configuration, and Filestore mounts.

Software Engineer

Feb 2025 - Nov 2025

Sogeti (on mission at BGL BNP Paribas) · Luxembourg

I developed and maintained internal banking applications for BGL BNP Paribas within an Agile tribe and squad.

- I delivered customer mandate management features with Spring Boot, batch processing, and Angular with Angular Material.
- I developed REST APIs and internal applications with Hibernate and PostgreSQL/Oracle, focusing on security and performance.
- I automated builds and deployments with Maven and Jenkins, and maintained code quality through unit tests, SonarQube, and peer reviews.

Selected company projects: BGL BNP Paribas - Mandate Management

Java Full-Stack Engineer

Sep 2024 - Jan 2025

CBTW · Luxembourg

I developed enterprise web applications and modernized Java services with Spring Boot, Angular, and PostgreSQL.

- I developed enterprise applications with Spring Boot, Angular, and PostgreSQL, refactoring monolithic components into microservices.
- I built REST APIs for service integration with attention to versioning, security, and documentation.
- I automated deployments with Jenkins and Docker, reducing manual release work.

Analyst Developer

Oct 2019 - Aug 2024

VERMEG for Banking and Insurance · Tunis, Tunisia

I progressed from Software Developer (Oct 2019 - Dec 2022) to Analyst Developer (Jan 2023 - Aug 2024), taking ownership of backend components and cross-team solution design for insurance and financial applications.

- I reduced API response time by 50% through query optimization and caching strategies.
- I built automated workflows using ActiveMQ for bulk processing, increasing efficiency by 40%.
- I owned core backend components and worked with business, architecture, and framework teams on requirements, microservices, and API contracts.

Selected company projects: SalesFlow · PackManager · Solife Digital & Generali Portal · Magikforms · Insurance Pricing Module

Personal projects

Slide Agent

Creator & Maintainer

Open-source presentation engine that turns AI-directed designs into editable PowerPoint decks.

- I built a TypeScript engine that compiles a grid-based composition language into native PowerPoint with editable charts, tables, theme colors, and embedded fonts.
- I implemented text measurement from font files, layout and contrast checks, and offline OOXML schema validation.

TypeScript · Node.js · MCP · OOXML · LLM APIs · Vitest · GitHub Actions

[slide-agent.ghassen.io](#) · [Source](#)

Pockito

Creator & Full-Stack Engineer

Personal and shared money management through mobile, web, and AI-agent interfaces backed by a shared domain service.

- I designed a Spring Boot backend with REST and MCP adapters over a shared domain service, alongside Flutter and Nuxt clients.
- I integrated Keycloak authentication, OAuth discovery for MCP clients, and event-driven notifications using Redis Streams.

Java 25 · Spring Boot · Flutter · Nuxt · PostgreSQL · Redis · Keycloak · Kubernetes · MCP

[pockito.ghassen.io](#)

SubMate

Creator & Maintainer

Chrome extension for synchronized subtitle translation on Netflix, TVer, and Prime Video, with on-device translation by default.

- I built a Manifest V3 extension using Chrome's on-device Translator API, with local caching and no SubMate backend, account, or telemetry.
- I designed platform adapters over a shared core for TTML/WebVTT parsing, TVer HLS segment assembly, and synchronized fullscreen overlays.

TypeScript · Chrome Extensions (MV3) · Translator API · LLM APIs · IndexedDB · Vitest

[submate.ghassen.io](#) · [Source](#)

Also built:

Orbit Ways: Browser-based multiplayer strategy board game with real-time game state.

Traffic Forward: Spring Boot service for temporary API routing, HTTP debugging, and request/response auditing.

Technical skills

Backend & Full-Stack Development: Java · Spring Boot · Helidon · MicroProfile · Hibernate · Angular · Vue.js · Nuxt · TypeScript · Node.js

Mobile & Browser: Flutter · Dart · Chrome Extensions (MV3)

AI & Agent Tooling: Model Context Protocol (MCP) · LLM API Integration

Database & Data Processing: PostgreSQL · Oracle (PL/SQL) · Redis

APIs & Messaging: RESTful API Design · OpenAPI · Kafka · ActiveMQ

DevOps & Infrastructure: Kubernetes · Google Cloud Platform (GCP) · Google Kubernetes Engine (GKE) · GitHub Actions · Docker · Jenkins

Quality, Security & Tools: JUnit · Mockito · SonarQube · Keycloak (OAuth2, OIDC, JWT) · JProfiler · Git

Education & certification

Engineer's Degree in Software Architecture Engineering · ESPRIT (Private Higher School of Engineering and Technology) · Sept 2019 - July 2023

Bachelor's Degree in Computer Science · ISIMM (Higher Institute of Computer Science and Mathematics) · Sept 2015 - June 2019

Oracle Certified Professional, Java SE 11 Developer · Oracle · Sept 2022

Languages

Arabic: Native · English: Advanced, Fluent · French: Advanced, Fluent · Japanese: Beginner, Basic Proficiency